

**BASICS OF INFECTION & IMMUNITY Module (BioMM - 2152)
for Anesthesia year II students**

- **Introduction to Medical Mycology**
- **The basics of host-pathogen interactions**
 - **Antimicrobial drugs**
 - **Sterilization and disinfection**

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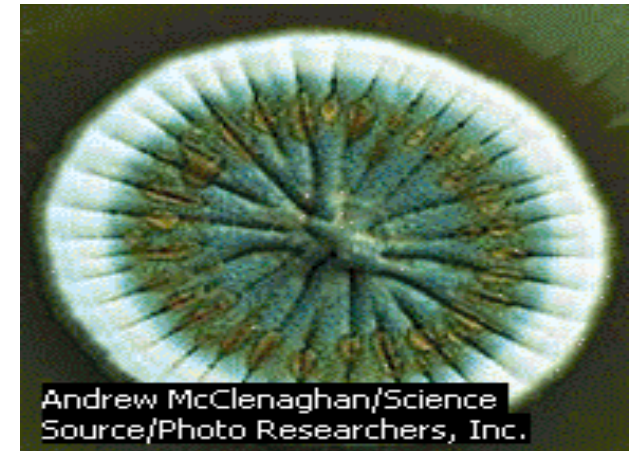
- **Introduction to Medical Mycology**

Outline of the lecture

- Introduction
 - General characteristics of fungi
- Basic fungal cell structure
- Morphological forms of microscopic fungi
- Basic lab methods for fungal diagnosis

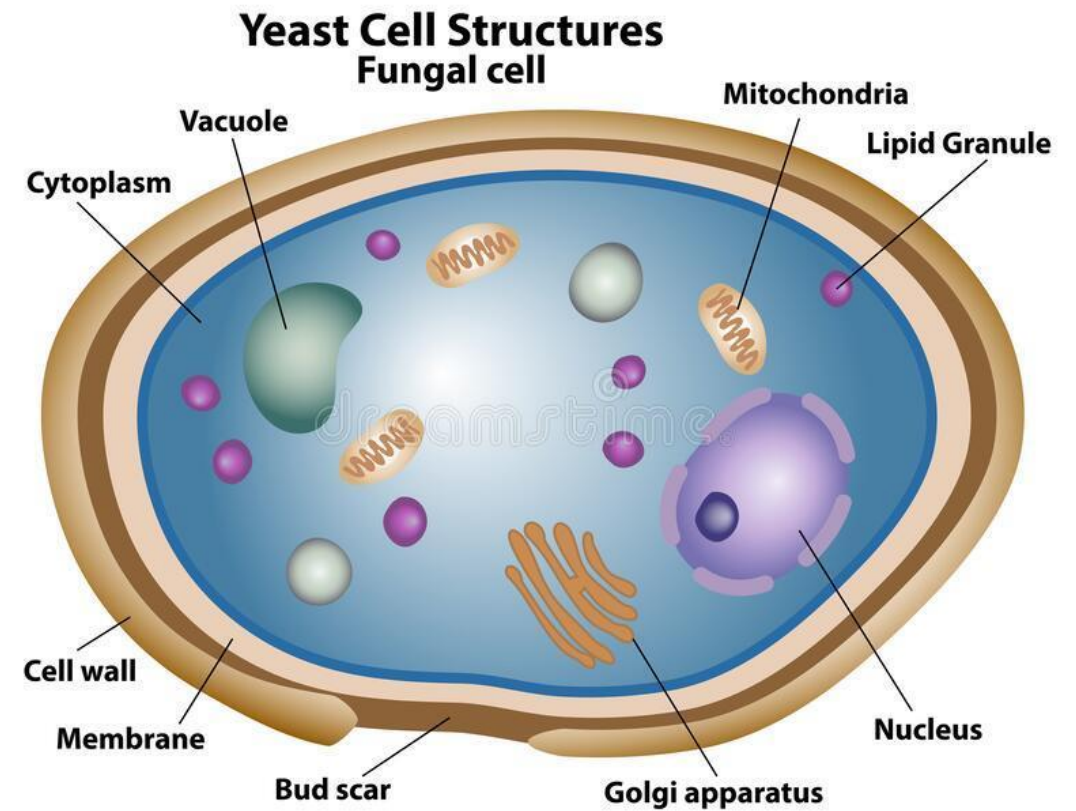
Introduction to fungi

- **Mykes** (Greek word): Mushroom
- Medical Mycology is the study of fungi that infect humans
- Fungi are eukaryotic organisms
 - Macroscopic and microscopic forms
 - Single celled or multicellular
- Very diverse groups (> 90,000 known species)
- Increasing trend of fungal infections
 - Only < 300 are pathogenic to humans
- Ubiquitous in the environment
 - Soil, plants, animals, sea water, foods
- Some are normal flora of humans
 - Candida species
- Non-motile



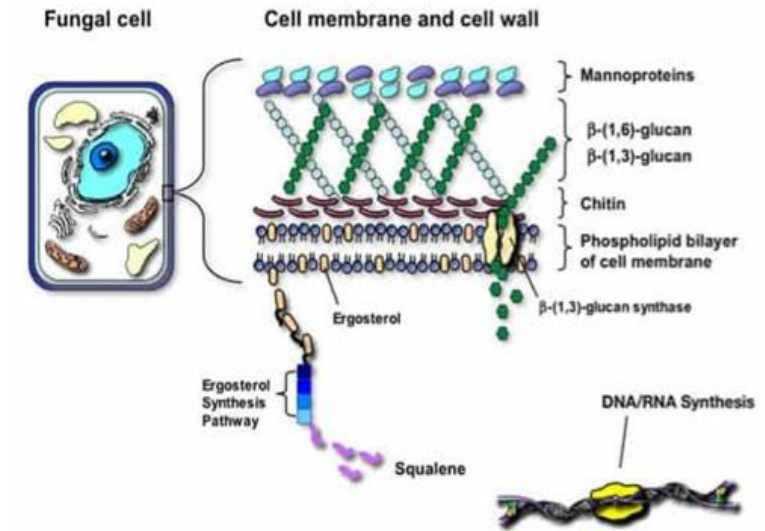
Fungi structure and morphology

- Fungal cell structure (eukaryotic)



- **Fungal cell structure**

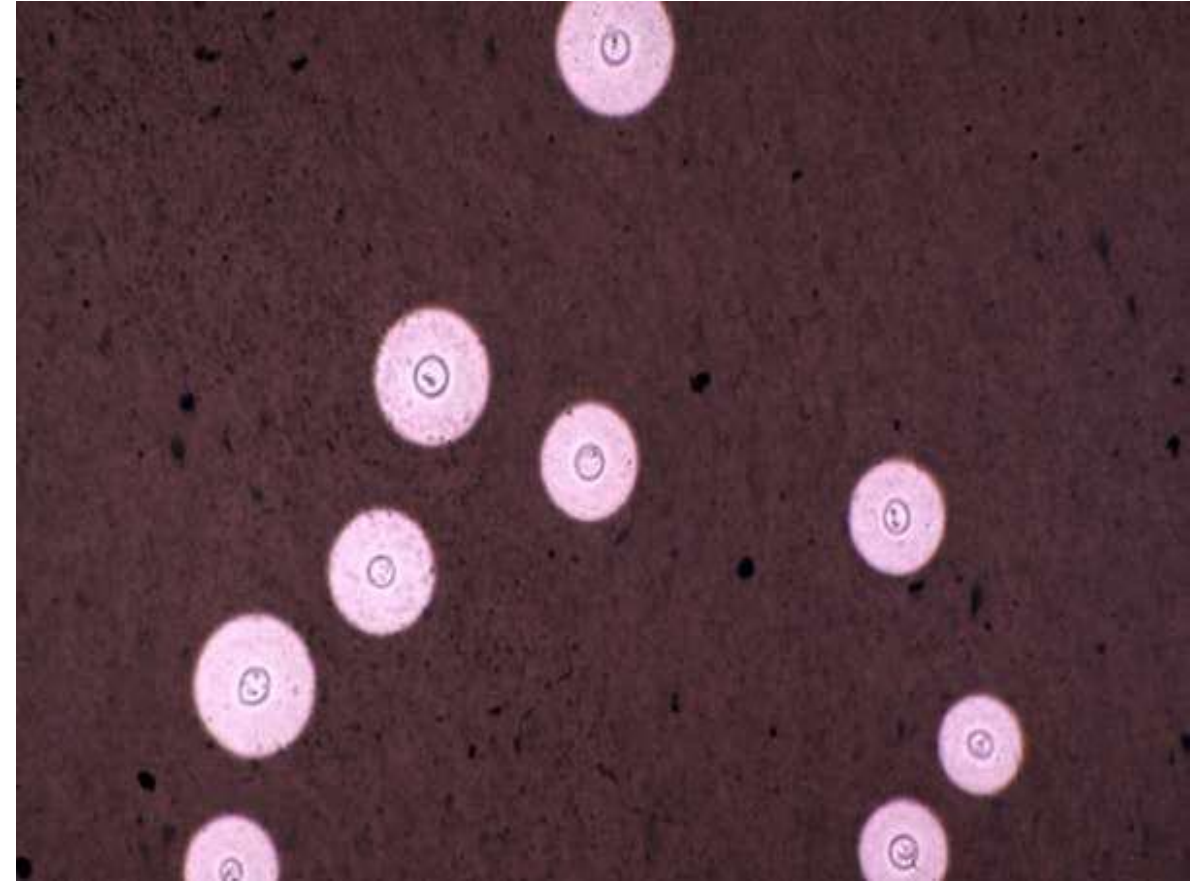
- Have cell wall made up of **chitin**
- Cell membrane consist of **ergosterol**



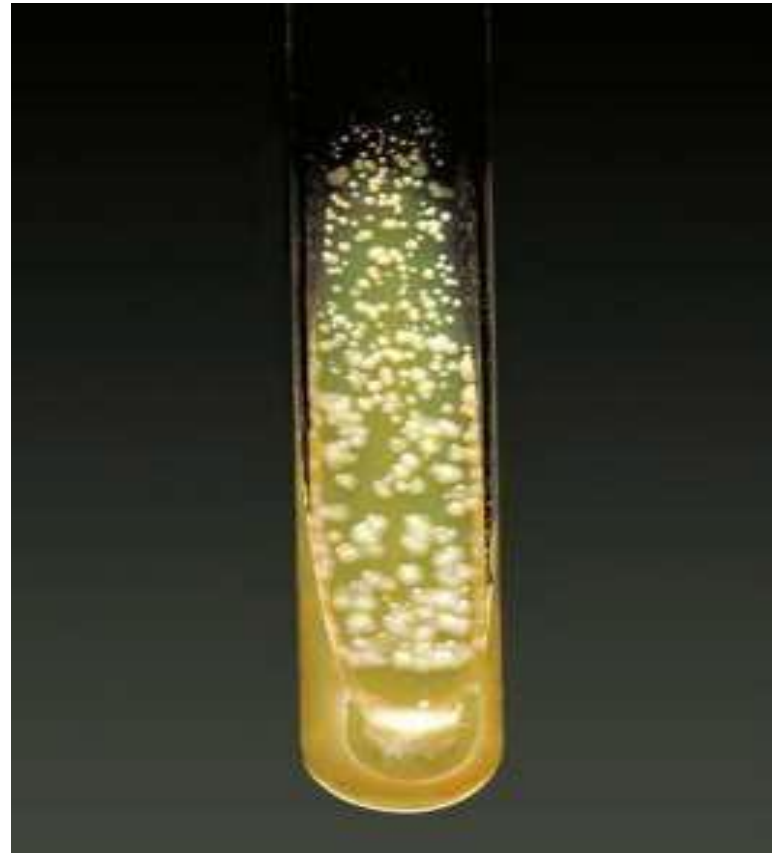
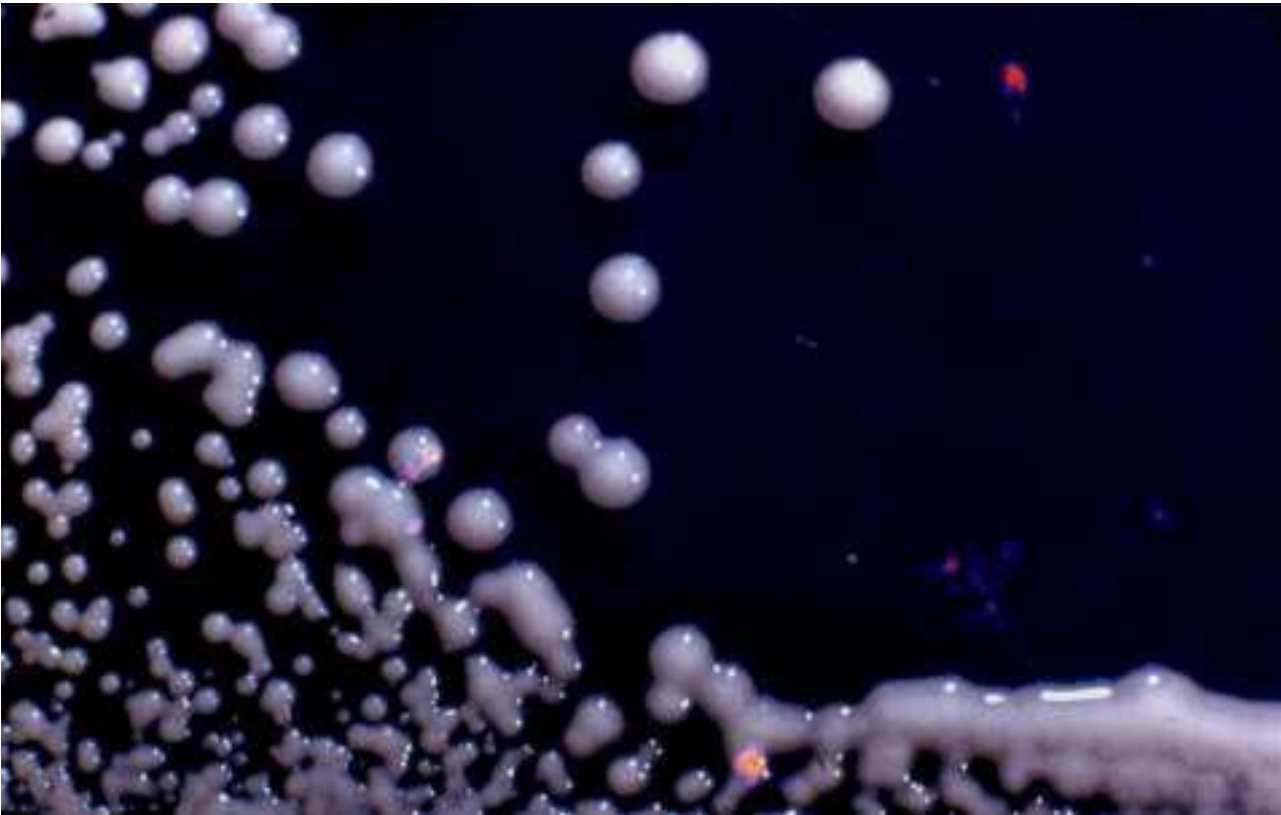
Morphological features of fungi

1. Yeasts

- Round or oval bodies
- Uni cellular fungi
- Produce smooth, creamy colonies on culture media
- Examples
 - *Saccharomyces cerevisiae* (brewer's and baker's yeast)
 - *Cryptococcus neoformans*

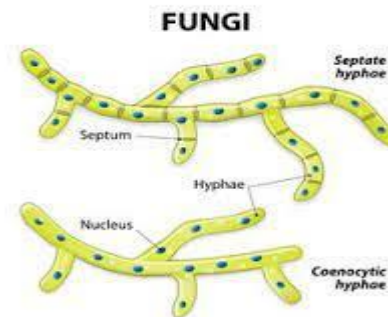
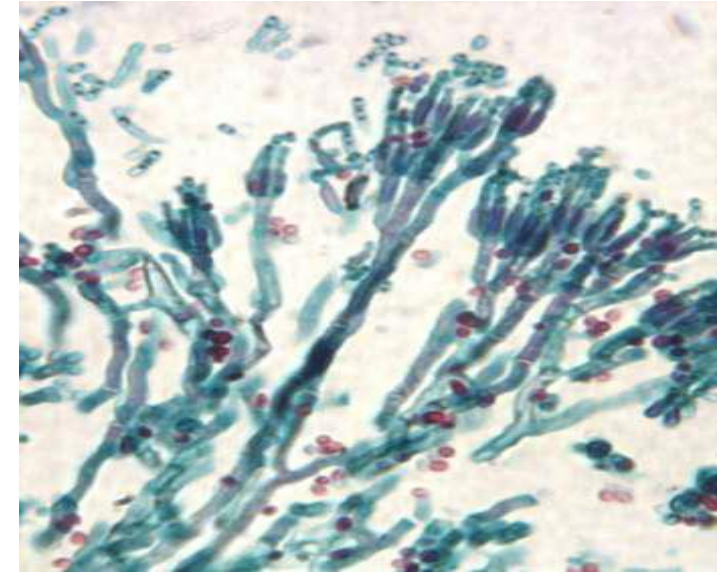


Yeast colonies



2. Molds

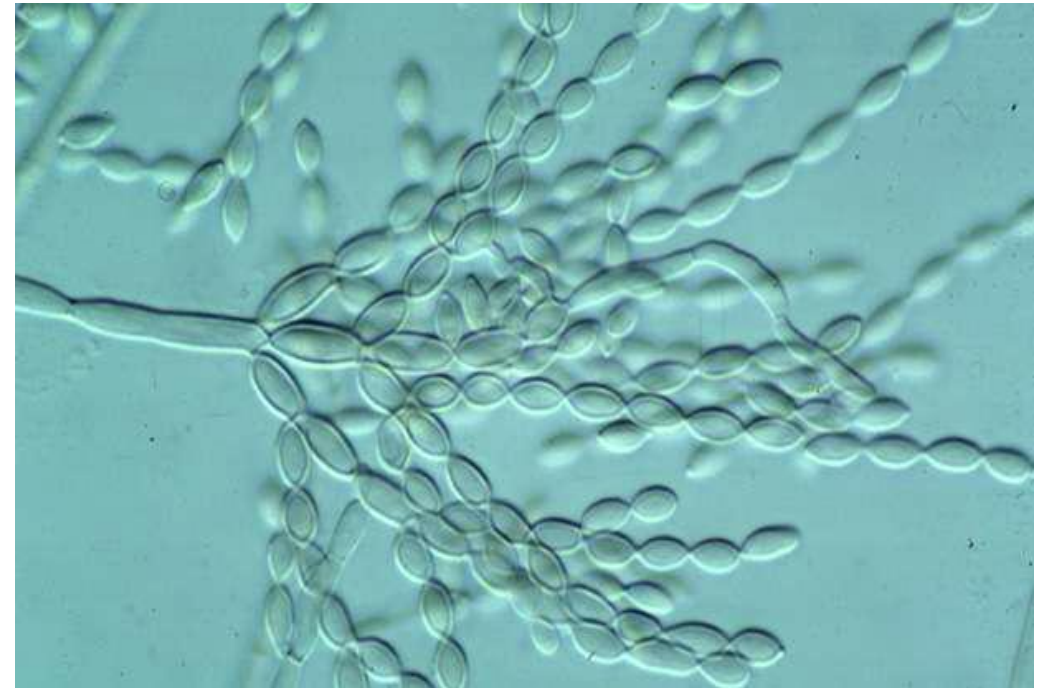
- Filamentous fungi
- Produce long branching filaments called hyphae
- Reproduce by the formation of different types of spores
- Colonies are strongly adherent to the medium/cannot be emulsified in water
- Example
 - Dermatophytes, Aspergillus,
 - Penicillium, Mucor, Rhizopus



3. Yeast like fungi

- Grow partly as yeasts and partly as elongated cells resembling hyphae which are called pseudo hyphae

e.g. *Candida albicans*

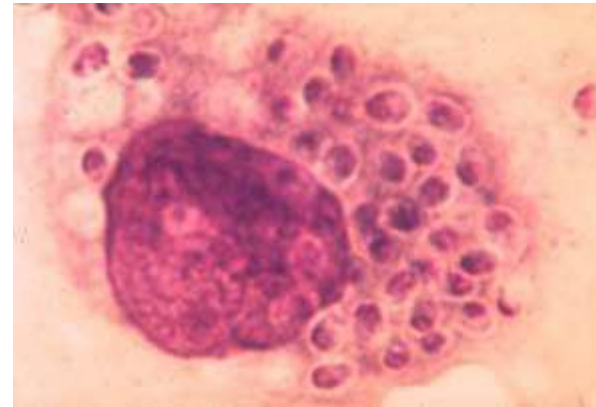


4. Dimorphic fungi

- Occur in 2 forms
- Molds (Filaments) – 25 oC (soil)
- Yeasts – 37 oC or in host tissue
- Most fungi causing systemic infections are dimorphic:
 - *Histoplasma capsulatum*
 - *Blastomyces dermatidis*
 - *Coccidioides immitis*
 - *Penicillium marneffei*



Hyphae (*Histoplasma capsulatum*)



Yeast within histiocyte (*Histoplasma capsulatum*)

Basic lab methods for fungal diagnosis

Introduction

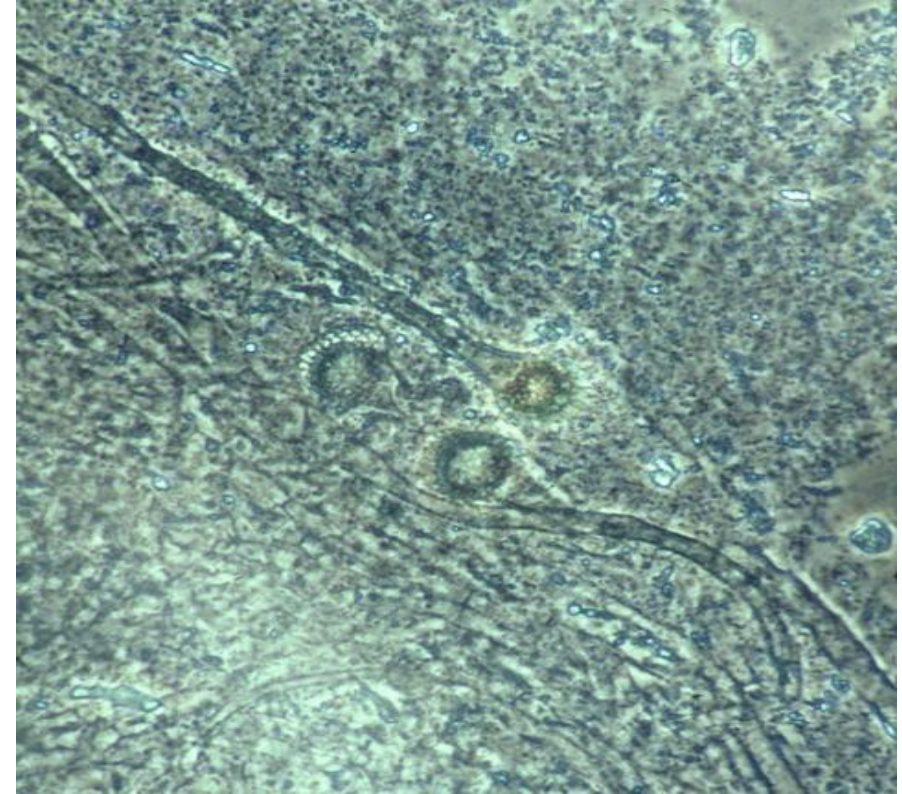
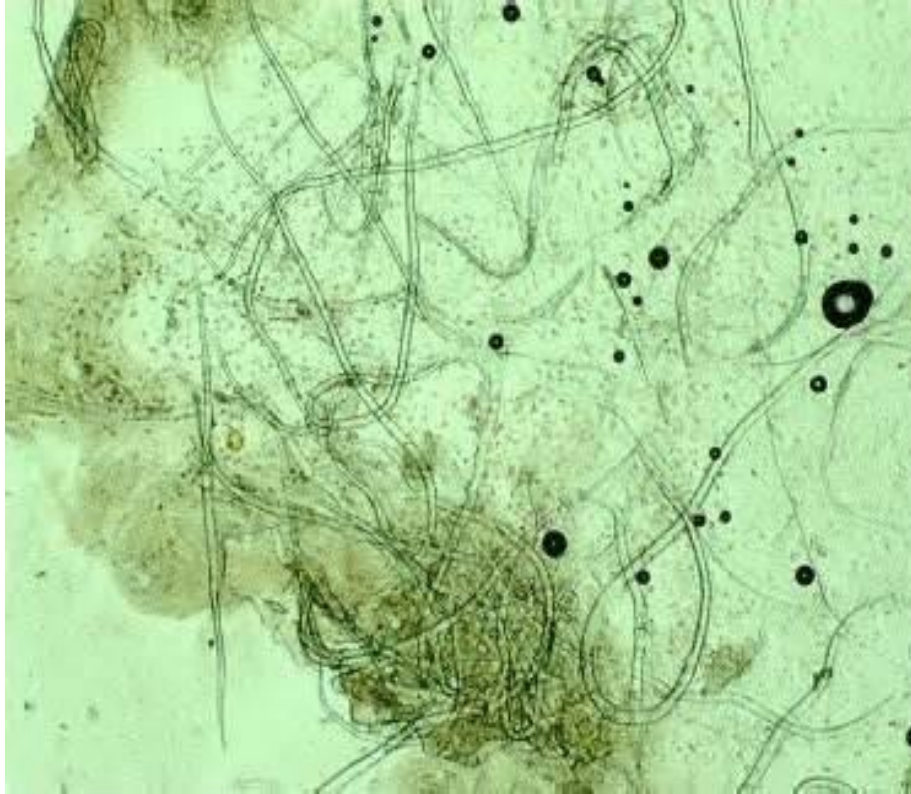
Types of fungal infections

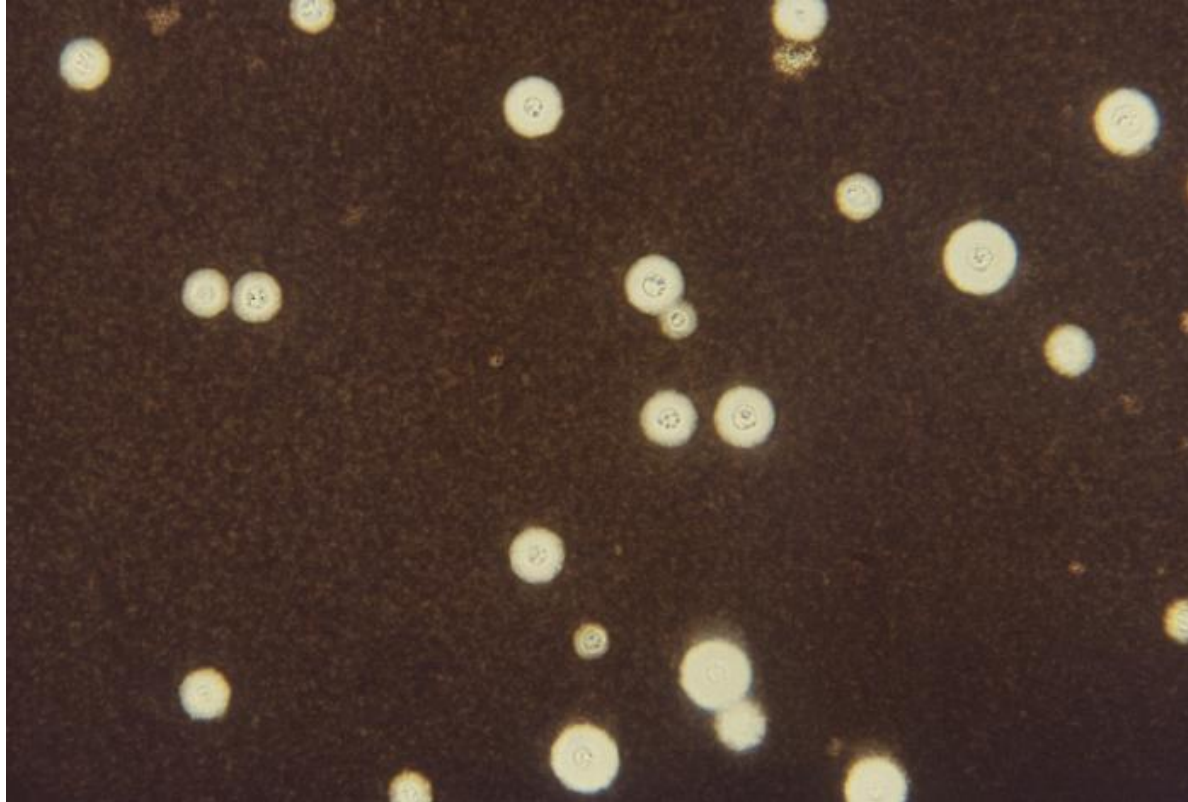
- **Superficial infections** - limited to the outer layers of the skin and hair
- **Cutaneous infections** - located deeper in the epidermis, hair and nails
- **Subcutaneous infections** - involve the dermis, subcutaneous tissues and muscle
- **Systemic** - generally originating in the lungs and other organs.
- **Opportunistic** - may involve a variety of body sites.

Basic lab methods for fungal diagnosis

- Direct examination
 - **Wet mount** - good for yeast; examination is done in natural environment, so loss of fragile structure is minimal.
 - **10% KOH** - done on skin scrapings, hair, nails, sputum, vaginal specimens.
 - **Gram staining**
 - **Gram Stain** - generally **fungi are gram positive**; but *Actinomyces* are gram variable.
 - **Other staining methods**
 - **India Ink**- for capsule of *Cryptococcus neoformans* in CSF specimens
 - **Lactophenol Cotton Blue** - very popular for quick evaluation of fungal structures; **stains chitin** in cell walls of fungi.
- **Fungal culture**
 - Sabouraud's Dextrose agar (SDA)
 - Brain Heart Infusion (BHI) medium
- Serological tests
- PCR & other molecular methods

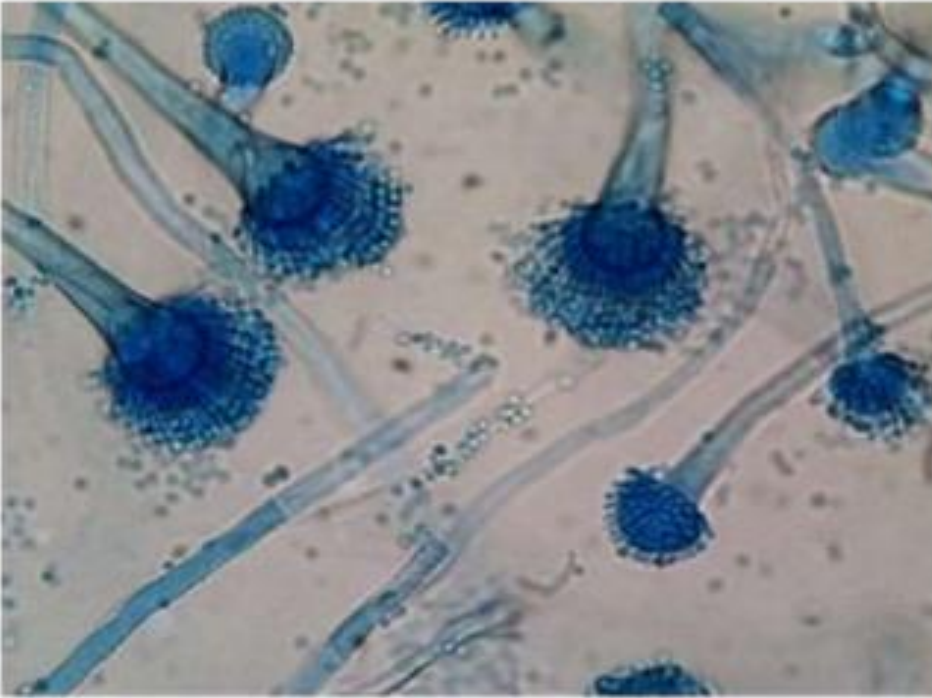
KOH positive for hyphae - culture is necessary to identify fungus





India Ink Stain showing capsules of *Cryptococcus neoformans*

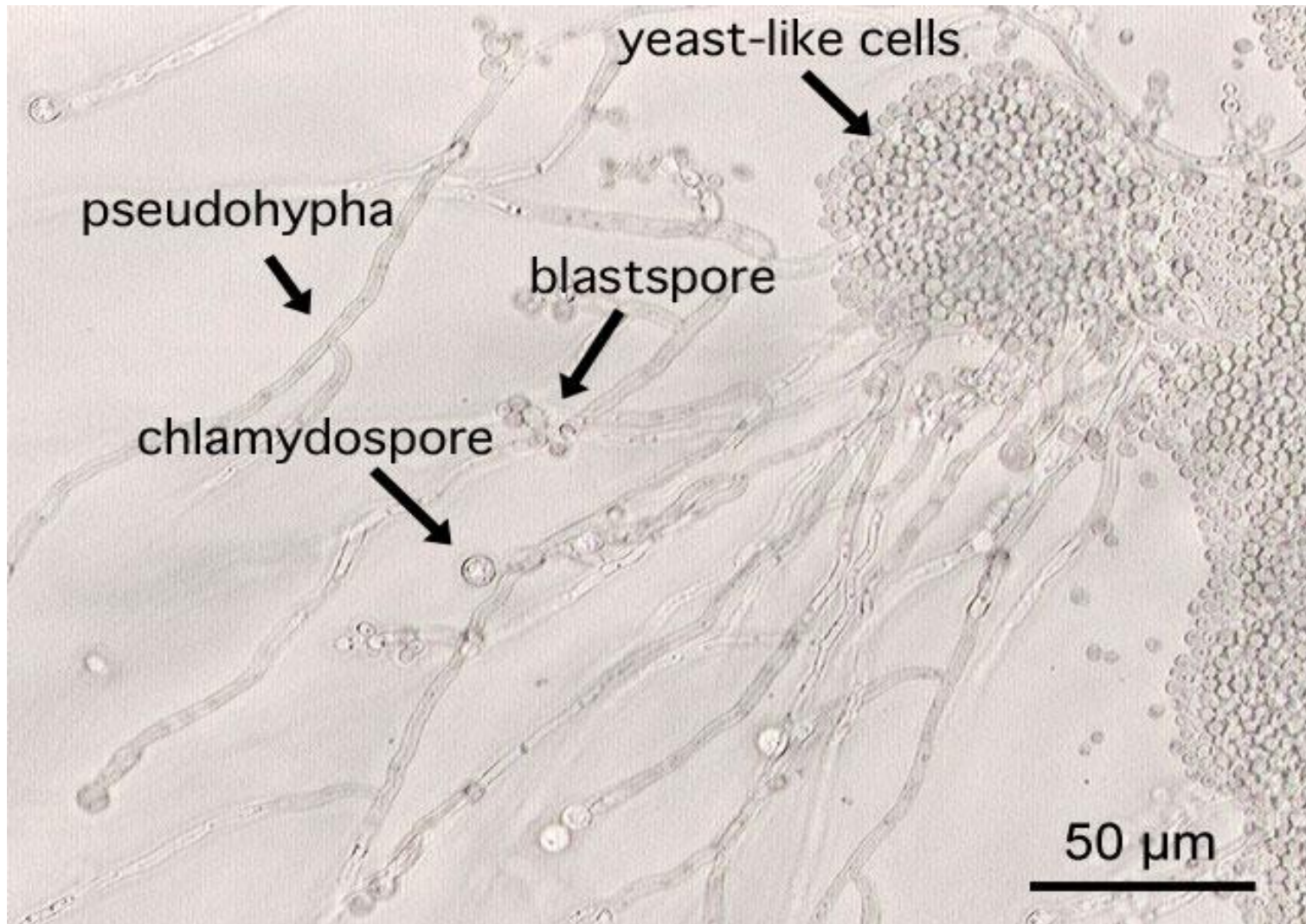
Pictures of fungi on LPCB mount



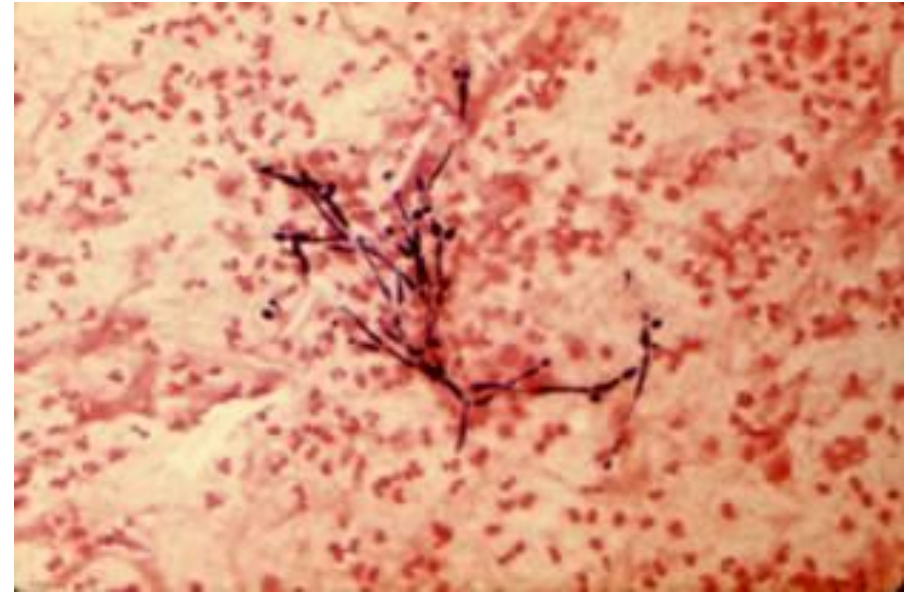
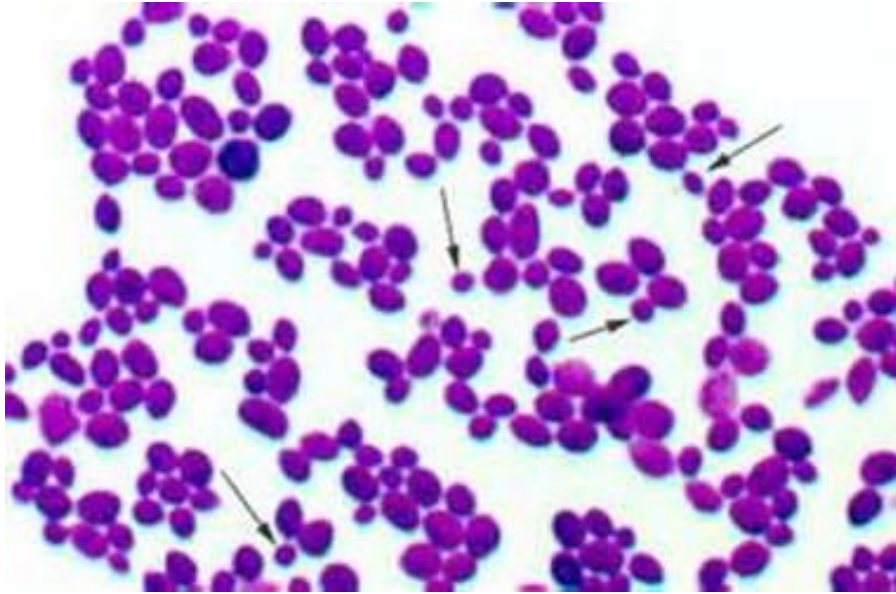
Aspergillus



Penicillium

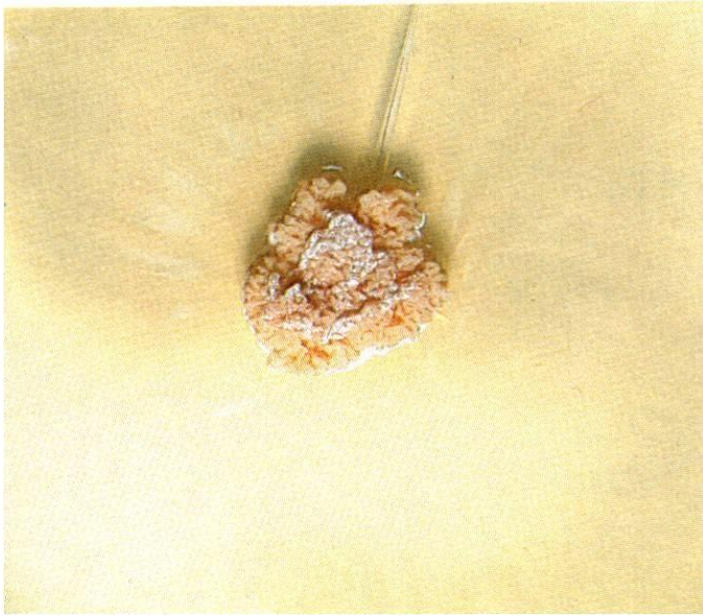


Saline wet mount -*Candida albicans*

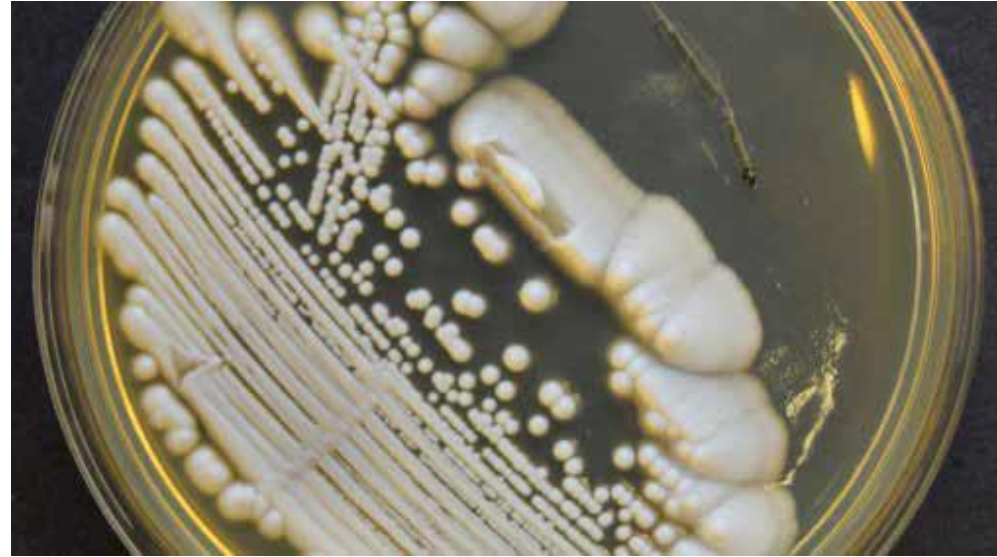


Gram stain - *Candida albicans*

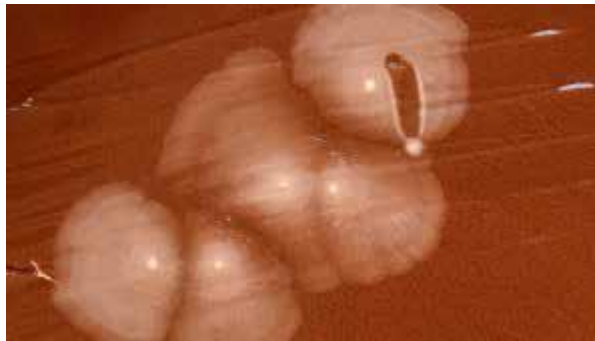
Fungal culture: Colony morphology (macroscopic features)



Verrucose



Umbonated colonies



Flat colonies

- **The basics of host-pathogen interactions**

Outline of the lecture

- Symbiotic Associations
- Normal Human Flora
- Mechanisms of Bacterial Pathogenicity

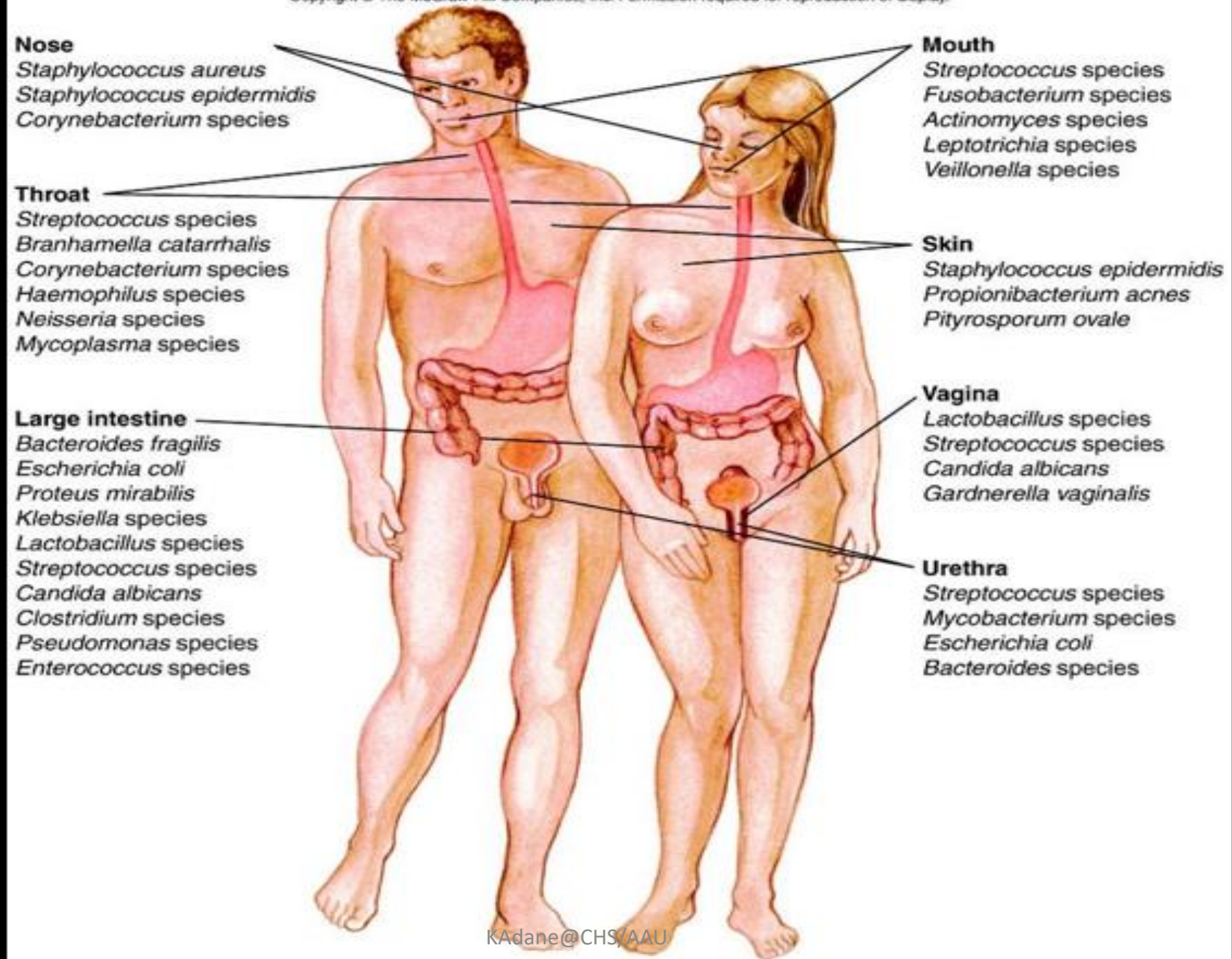
Introduction

Symbiotic Associations

- **Mutualism** - both members of the association benefit
 - Lactic acid bacteria that live on the vaginal epithelium
- **Commensalism** - no apparent benefit or harm to either member of the association.
 - *Staphylococcus epidermidis* on the skin
- **Parasitism** - one member of the association lives at the expense of the other member
 - Pathogenic bacteria

Normal Human Flora

- **A mixture of organisms** regularly found at any anatomical site (mostly on skin and mucous membranes)
- Only certain body parts are free of microorganisms
 - Blood, brain, muscle, CSF, lungs, bladder
- Colonization occurs shortly after birth
- Mainly includes bacteria & fungi
- More bacterial normal flora than human cells in the body
 - 10^{13} body cells vs. 10^{14} bacteria



Types of normal flora

- **The resident flora**

- Relatively fixed at a given body site
- If disturbed, they promptly reestablish themselves

- **The transient flora**

- Inhabit a given body site for only hours, days, or weeks
- If disturbed, they do not reestablish themselves

Benefits of normal flora

- **Synthesizes and excretes vitamins**
 - Enteric bacteria secrete Vitamin K and Vitamin B12
- **Prevents colonization by pathogens**
 - Compete for attachment sites or for essential nutrients
- **May antagonize other bacteria**
 - Intestinal bacteria produce **bacteriocins** that kill other bacteria
- **Stimulates the production of cross-reactive antibodies**
 - Antibodies produced against normal flora may protect infection

Could normal flora be harmful ?

- May cause disease under certain circumstances
 - If forcefully removed from the environment they adapted- **dysbiosis**
 - Viridians Streptococci may cause **infective endocarditis** following tooth extraction
- If the conditions of the host is changed
 - Changes to the biology of mouth may lead to **dental caries**

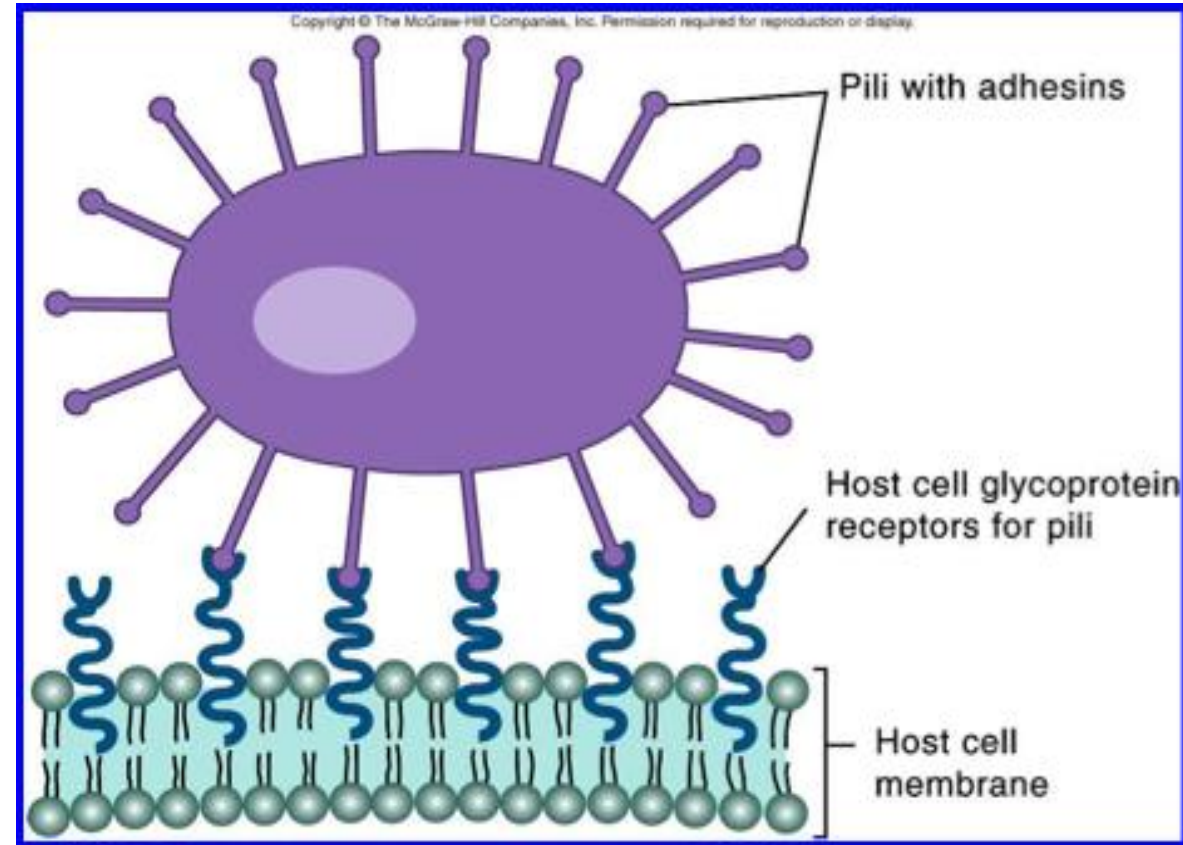
General Mechanisms of Bacterial Pathogenicity

- Two mechanisms
 - The ability to invade tissues: **invasiveness**
 - The ability to produce toxins: **toxigenesis**

1. Invasiveness

A. Colonization: first stage of microbial infection

- Adherence requires the participation of two factors
 - Adhesin and a receptor



B. Invasion

- Facilitated by invasins
 - Proteins/enzymes that act locally to damage host cells
 - Facilitate the spread of the pathogens
- Examples of invasins
 - **Hyaluronidase**- hydrolyzes hyaluronic acid
 - Produced by **Staphylococci, Streptococcus, & Clostridia**
 - **Collagenase**- breaks down collagen
 - Produced by **Clostridia**
 - **Streptokinase** - plasminogen to plasmin
 - Produced by **Staphylococci, Streptococci**

2) Toxigenesis

- Over 220 known bacterial toxins
 - Toxin could be ingested following production (food poisoning)
 - Toxin could be produced following colonization
- Can be divided into two groups
 - Exotoxins
 - Endotoxins

Exotoxins

- Excreted by living cells
- Produced by both gram +ve and gram-ve bacteria
- Controlled by extrachromosomal genes (**plasmids**)
- Exotoxins could be divided into 3 main groups
- **Neurotoxins** => affect neurons
 - *Botulinum toxin* - neurotoxin of *C. botulinum* leads to flaccid paralysis
- **Enterotoxins** => affect the gastrointestinal
- **Cytotoxins** => damage cells
- **Erythrogenic toxin**-cytotoxins of *S.pyogenes*
 - Damage blood capillaries and produce red skin rash

Endotoxins

- Integral part of cell wall of gram-negative bacteria
 - Released on death and in part during growth
 - Found only in gram-negative bacteria (LPS part)
 - Synthesis directed by chromosomal genes
- **Biologic effects of endotoxin**
 - Induce fever due to release inflammatory cytokines
 - May cause disseminated intravascular coagulation

- Antimicrobial drugs

Outline of the lecture

- Antimicrobial drugs
- How do antimicrobial drugs work?
- Mechanisms of Action of Antibacterial Drugs
- Antimicrobial resistance (AMR)

Antimicrobial drugs

- Chemical substances of natural or synthetic origin to treat infections
 - **Antibiotics**
 - Natural substances derived from microorganisms
 - **Semi-synthetic antibiotics**
 - Partially modified in the laboratory
 - **Synthetic drugs**
 - Man-made chemicals

Some Antibiotics: Microbial Sources

Bacteria	Fungi
<i>Micromonospora</i> spp. Gentamicin	<i>Penicillium</i> spp. Penicillin Griseofulvin
<i>Bacillus</i> spp. Polymyxins Bacitracin	<i>Cephalosporium</i> spp. Cephalosporins
<i>Streptomyces</i> spp. Streptomycin Tetracyclines Erythromycin Kanamycin Rifampin Neomycin Nystatin	

How do antimicrobial drugs work?

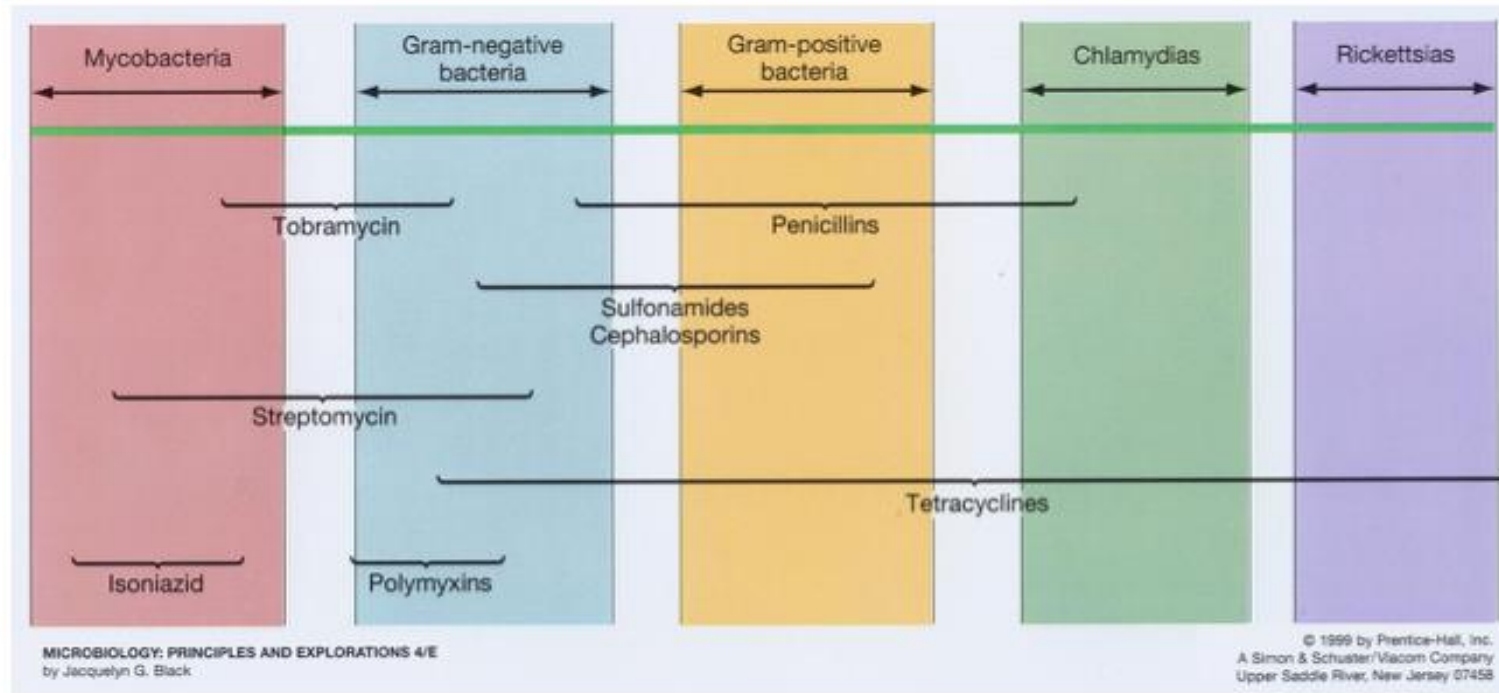
- **Selective toxicity**
 - Ability to harm the target microbe without harming the host
- Discuss the following in terms of their selective toxicity
 - Antibacterials
 - Antifungals
 - Antivirals

Antibacterial drugs

The Action of Antibacterial Drugs

- Bactericidal
 - Kill microbes directly
- Bacteriostatic
 - Prevent microbes from growing

Antibacterial drugs

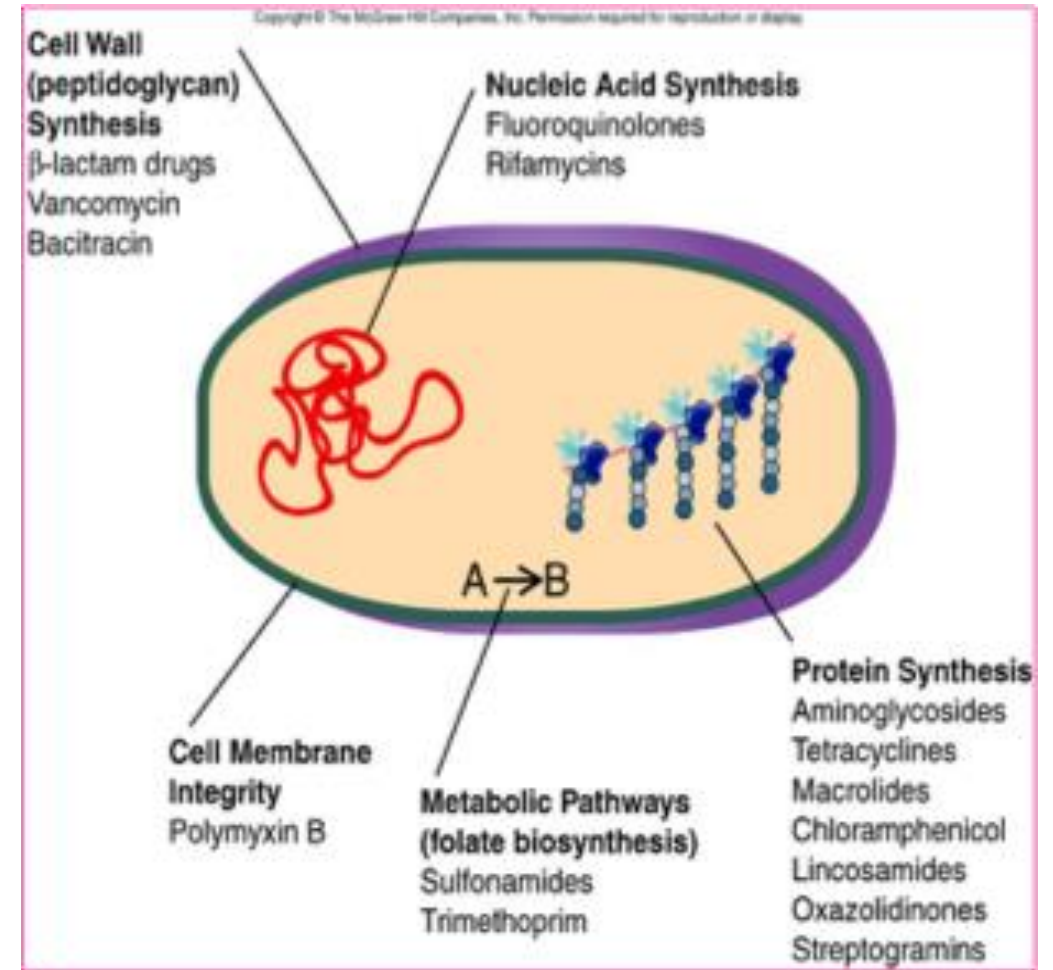


Narrow-Spectrum Antimicrobials vs Wide-Spectrum Antimicrobials

Antibacterial drugs

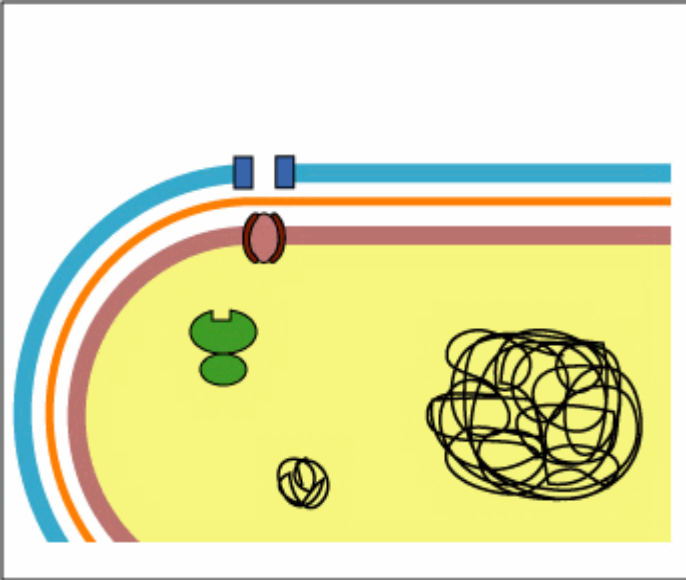
Mechanism of actions

- Inhibition of cell wall synthesis
- Interference with cell membrane integrity
- Inhibition of protein synthesis
- Inhibition of nucleic acid synthesis
- Inhibition of metabolic pathways



Antimicrobial resistance (AMR)

- One of the current most serious public health threats
- **Basis of microbial resistance**
 - Inherent resistance
 - Reduced permeability in some Gram-ve bacteria
 - Lack of target site
 - **Acquired resistance**
 - Mutation and selection
 - Exchange of genes



Antimicrobial resistance

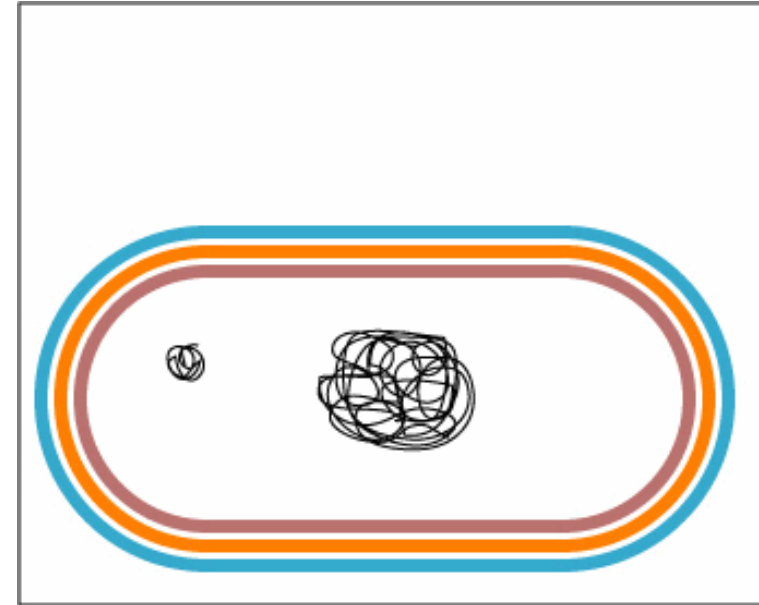
ability

and efflux of a drug

- Inactivation of a drug
- Modification of a drug target

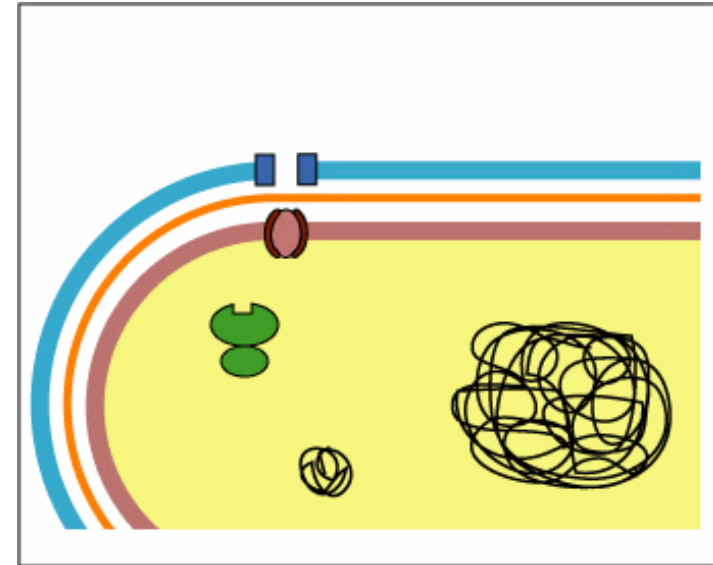
- **Inactivation**

- B-lactamase
- Chloramphenicol acetyl transferase



- **Altered target site**

- Penicillin binding proteins (penicillins)
- RNA polymerase (rifampin)
- 30S ribosome (streptomycin)



Factors contributing to Antibiotic Resistance

- **Environmental factors**
 - Widespread use of antibiotics in animal husbandry and agriculture
 - Huge populations and overcrowding
- **Drug Related factors**
 - Fake drugs
 - Over the counter availability of antimicrobials
 - Poor quality of the drug
- **Patient Related Factors**
 - Poor adherence
 - Self-medication
 - Misconception
- **Physician Related Factors**
 - Increased empiric use
 - Overuse of antimicrobials
 - Inadequate dosing

How to prevent AMR?

- Apply antimicrobial stewardship
 - Coordinated program that promotes the appropriate use of antimicrobials
- Work against all the factors listed

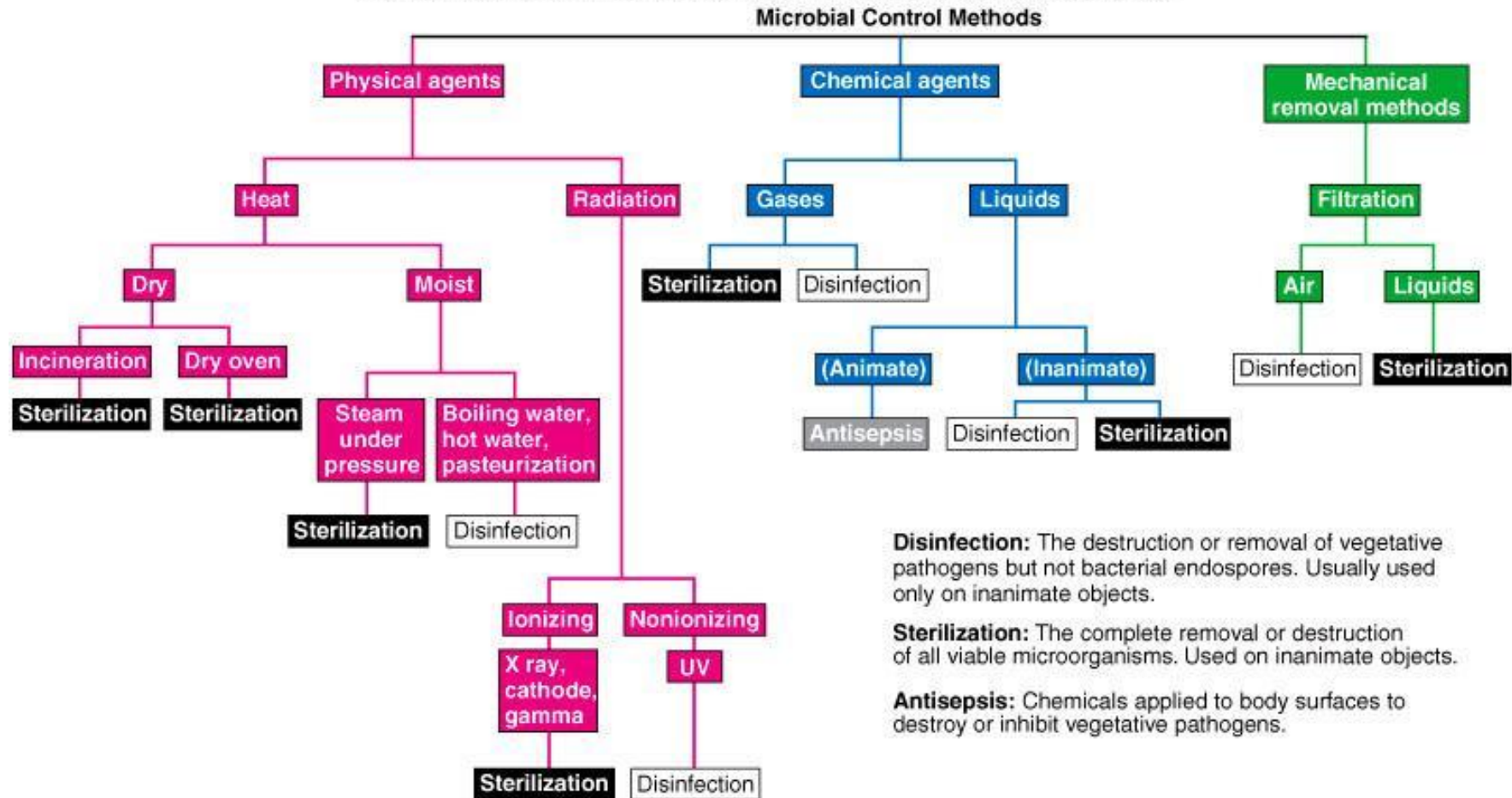
- **Sterilization and disinfection**

Outline of the lecture

- Sterilization and disinfection : Overview
- Methods of sterilization
- Disinfection

Sterilization and disinfection : Overview

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Introduction

- **Sterilization**

- A physical or chemical process that completely destroys or removes all microbial life, including spores.

- **Disinfection**

- Killing or removing of harmful microorganisms

- **Disinfectant**

- Products used to kill microorganisms on inanimate objects or surfaces

- **Antiseptic**

- A product that destroys or inhibits the growth of microorganisms in or on living tissue

Methods of sterilization

1. Physical methods

- Heat
 - Dry
 - Moist
- Radiation
 - U.V. light
 - Ionizing radiation
- Filtration

2. Chemical methods

Physical methods

- **Moist heat sterilization**

- Autoclaving(Steam under. pressure)
 - Uses hot water
 - Kills microorganisms by denaturing proteins
 - Done at 121 °C (250 °F) 15 pounds pressure for 15–20 minutes
-
- Most commonly used effective method of sterilization



Physical methods

- **Dry heat sterilization**
 - Sterilization using heat
 - Done using hot air oven
- Could be done
 - 170°C for 1 hour or
 - 160°C for 2 hours
- Can be used to sterilize glasses and sharp materials



Chemical sterilization

- Useful for heat sensitive materials
 - Plastics and lensed instruments, endoscopes
- Ethylene Oxide Chamber
- Activated alkaline Glutaraldehyde 2%

- **Disinfection**
 - Elimination of many or all pathogenic microorganisms, with the exception of bacterial spores using antiseptics and disinfectants
- Common disinfectants
 - Chlorine and hypochlorites
 - Hexachlorophene
 - Aldehydes
 - glutaraldehyde
- **Common antiseptics**
 - Alcohols (Ethyl alcohol, Isopropyl alcohol)
 - Alcohols have optimal activity when diluted 60 – 90%

Thank you!